

# Ayman Ismaili

## AI/ML Engineer/Data Scientist/Data Engineer

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Languages: Arabic (Native), French (Advanced), English (Fluent)

### Professional Summary

AI and Data Science engineering student at ENSAM-Meknès, passionate about transforming complex data into intelligent, high-impact solutions. My expertise covers Machine Learning, Computer Vision, and MLOps automation, with a strong focus on building scalable and efficient systems. I am currently seeking a final-year internship (PFE) starting July 2025, where I can contribute to innovative projects and apply my technical and analytical skills in a challenging environment.

### Education

- **ENSAM-Meknès, Morocco** Sep 2021 – Present  
*Engineering Degree - Industrial Engineering, AI & Data Science*

### Professional Experience

- **Machine learning Intern – GEOFLY** Jul – Sep 2025
  - Developed a lightweight UNet segmentation model achieving 83.34% IoU and 312 FPS.
  - Designed GIS-compatible polyline output with automated GeoJSON export.
  - Implemented RESTful API using FastAPI and an interactive frontend via Flask.
  - Containerized the entire application with Docker Compose.
  - Enhanced image preprocessing using contour detection and data augmentation.
  - Optimized GPU inference for real-time performance (382K params, 1.5MB model size).
- **Machine Learning Intern – Saiket System** Mar – May 2025
  - Built a customer churn prediction model with 70% precision and 80% recall.
  - Developed ETL pipelines and automated CI/CD workflows using Docker and GitHub Actions.
  - Engineered domain-specific features and applied cross-validation for robust results.
  - Tracked experiments and metrics efficiently using MLflow.
- **Engineering Intern (Observatory) – APTIV Morocco** Jul – Aug 2024
  - Observed and documented Lean Six Sigma methodologies and manufacturing processes.

### Technical Skills

- **Programming:** Python, Java, C++, Bash
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, Matplotlib, NLP, RAG, LLM, LLM finetuning
- **Computer Vision:** OpenCV, Pillow, Image Processing, Contour Detection, Polyline Extraction
- **MLOps & Cloud:** Docker, Kubernetes (basics), Terraform (IaC), FastAPI, Flask, CI/CD, Airflow, MLflow, AWS (Cloud)
- **Big Data:** PySpark, Spark MLlib, ETL, Feature Engineering, SQL (MySQL)
- **NLP:** LangChain, Ollama
- **Tools:** Git, GitHub, Jupyter
- **Methodologies:** Lean Six Sigma, Agile
- **Soft Skills:** Teamwork, Communication, Problem-solving, Adaptability

### Technical Projects

- **Automated ML Pipelines with Apache Airflow** *Python, Airflow, Docker – 2025*
  - Built end-to-end ML workflows using Airflow DAGs for data ingestion, preprocessing, model training, and evaluation.
  - Implemented dynamic task dependencies, XComs, and scheduling for efficient execution.
  - Dockerized the Airflow environment and monitored performance via Airflow UI.
  - Automated error handling and logging for reproducible ML pipelines.
- **CIFAR-10 Classifier Web App** *PyTorch, Streamlit – 2025*
  - Designed and trained a CNN model for CIFAR-10 classification.
  - Deployed a Streamlit web app with live training visualization and confusion matrix.
  - Published the application online for public access: [link](#).
- **Large-Scale Customer Segmentation** *PySpark, MLflow – 2025*
  - Processed 100k+ transaction records using PySpark for scalable data analytics.
  - Applied clustering with Spark MLlib and evaluated performance metrics.
  - Logged and compared model runs using MLflow for experiment management.
- **Scholarship Assistant Chatbot** *LangChain, Streamlit – 2024*
  - Developed a multilingual chatbot capable of answering scholarship-related queries.
  - Integrated LangChain for NLP pipeline and deployed the interface via Streamlit.
- **Flower Species Recognition System** *Keras, TensorFlow – 2023*
  - Built a CNN-based classifier for flower species recognition.
  - Visualized training metrics and validated performance on test data.

### Certifications

- AWS Cloud Practitioner Essentials Certificate – AWS Training and Certification (2025)
- Completed Python Machine Learning Course – Udemy (Apr 2025)
- Earned Six Sigma Yellow Belt – ECLEE (Apr 2025)
- Completed Python for Data Science – Udemy (Mar 2025)